IBM HR Analytics — Predict Employee Attrition

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Abstract: This project analyzes employee attrition using the IBM HR Analytics dataset containing 1,470 employee records and 35 features. The objective was to identify factors influencing employee resignations and build a predictive machine learning model to help HR proactively retain employees. Logistic Regression with SMOTE achieved 87% accuracy and 54% recall. Key attrition drivers identified include overtime burden, salary dissatisfaction, and poor work-life balance.

Introduction: Employee attrition leads to productivity loss, recruitment costs, and operational disruptions. This project aims to answer the business question: *"Which employees are most likely to leave, and why?"* Using data analytics, visualization, and machine learning, the project provides actionable insights for HR departments to improve employee retention strategies.

Tools & Technologies

Tool	Purpose
Python	Programming & Analysis
Pandas	Data Cleaning & Transformation
Seaborn & Matplotlib	Data Visualization
Scikit-learn	Machine Learning
SMOTE	Handling Class Imbalance
SHAP	Model Explainability
Power BI	Interactive Dashboard

Methodology:
1. Loaded and explored the IBM HR Analytics dataset.
2. Removed constant and non-predictive columns.
3. Encoded categorical variables and scaled numerical features.
4. Performed Exploratory Data Analysis (EDA) to identify attrition trends.
5. Applied SMOTE to balance the dataset.
6. Trained multiple classification models and selected Logistic Regression based on recall performance.
7. Used SHAP values to explain model predictions.
8. Developed an interactive Power BI dashboard and prediction application.

Key EDA Insights:
- Employees working overtime were significantly more likely to leave.
- Employees with lower salaries showed higher attrition rates.
- Employees aged 20–25 experienced the highest attrition.
- Poor work-life balance strongly correlated with resignations.
- Overtime had the strongest positive correlation with attrition.

Model Performance:
The following machine learning models were compared:

Model	Accuracy	Recall
Logistic Regression + SMOTE	87%	54%
Decision Tree + SMOTE	77%	44%
Random Forest + SMOTE	88%	28%
SVM + SMOTE	88%	51%

Conclusion: The project successfully identified major factors contributing to employee attrition and built a predictive model to support proactive HR interventions. Overtime burden, low salary, and poor work-life balance emerged as the strongest drivers of attrition. The developed solution enables organizations to identify at-risk employees early and implement targeted retention strategies such as compensation improvement, flexible work policies, and mentorship programs.

Future Enhancements:
- Real-time integration with HR systems
- Periodic model retraining
- Inclusion of employee feedback and sentiment analysis
- Enhanced AI-powered retention recommendation system